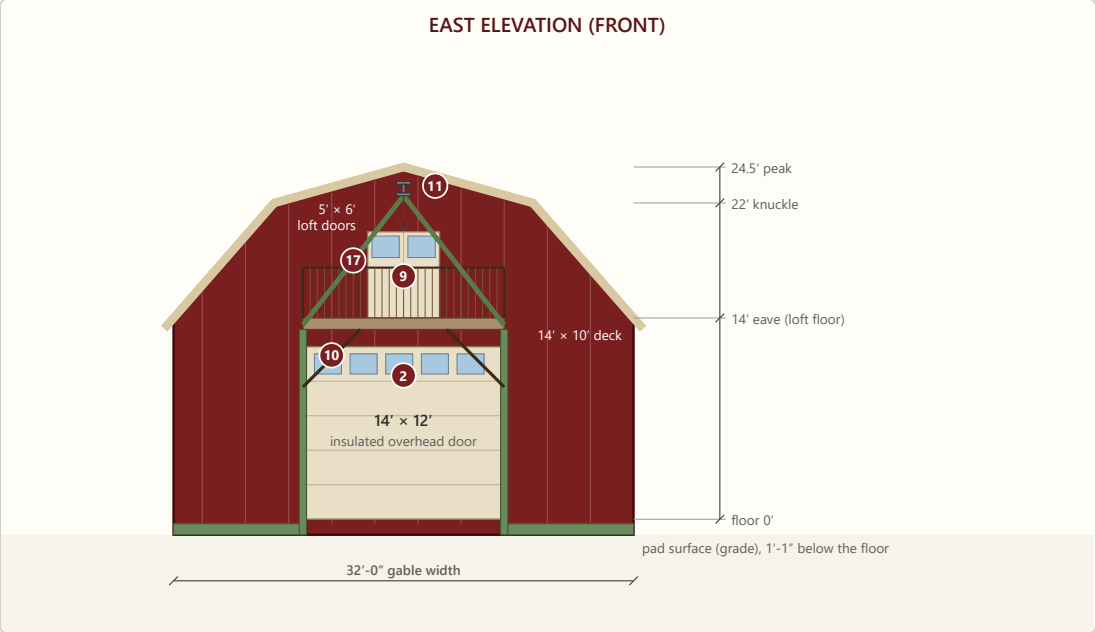
Workshop Pole Barn

Mountain Prairie Acres · 32 × 48 ft gambrel pole barn: RV garage, workshop, and storage loft · **PLANNED FOR 2028** · built with family

Every drawing, count, and price in this package is computed from one design model at the planned footprint in the site data. County rules sets the building against what Elmore County's ordinance says.

> **Planned for 2028. Nothing is ordered, permitted, or built.** These are planning drawings and figures. Have Elmore County and an Idaho-licensed engineer confirm every load and dimension, and get quotes, before anything is ordered. Prices are 2026 ballpark figures.

The Building in Brief



East elevation, the front: the 32′ × 48′ building seen from the driveway side. Every picture in the package is drawn from one design model, so the length lives in one place.

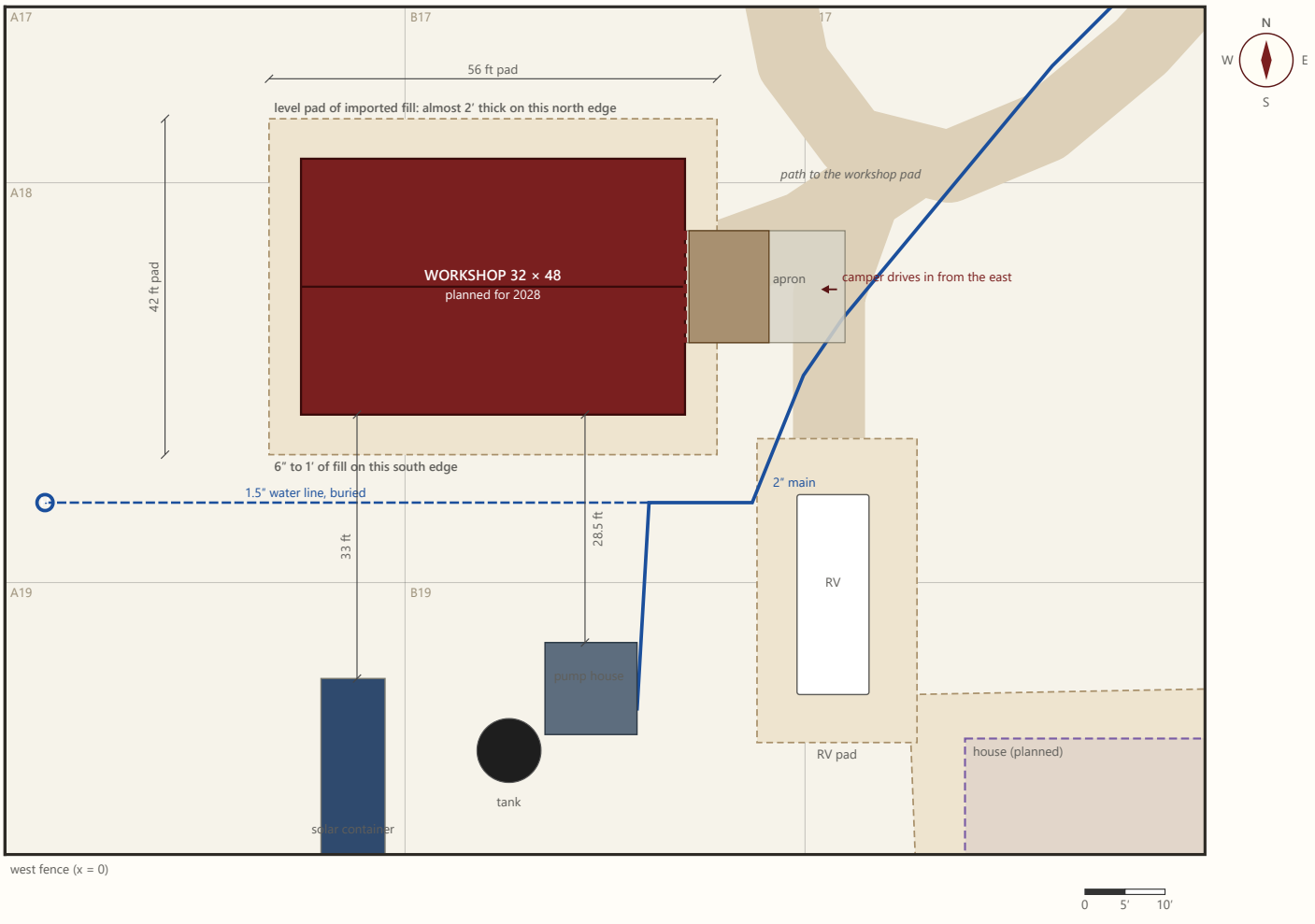
Use	RV garage, workshop, and storage loft. What the building really holds decides its permit path: see County rules
Camper	2024 Forest River Rockwood Mini Lite 2517S towable, 25′-11″ × 8′ × 10′-11″ (length × width × height, the manufacturer's listed exterior size); 5,774 lb dry, 7,615 lb GVWR (fully loaded). The first story is sized to drive it in: the door is 12 ft tall and there is 13′-2″ clear under the loft floor
Footprint	32 × 48 ft: 1,536 sq ft on the main floor and the same on the loft floor, of which about 950 sq ft has 7 ft or more of headroom (19.75 ft wide down the middle)
Bays	Sidewall posts at 8 ft: 8-8-8-8-8-8, with 19 piers and 25 trusses at 2 ft OC. The length is a whole number of 8 ft bays, so the posts, the 16 ft girts, the 4×8 sheets, and the 36 in steel panels land on a clean module
Wall framing	2×6 girts laid flat on the outside face of the posts at 24 in OC, screwed to every post: 7 rows on each wall, the lowest pressure-treated. The boards are 16 ft, two bays, so every joint falls on a post; alternate rows start with an 8 ft piece so neighboring rows never join over the same post, and 18 ft boards cut in half fill the 9 ft runs beside the east door. The sheathing goes over the girts, so the wall is about 10″ thick (wall framing sheets)
Heights	14 ft eave (the loft floor), 22 ft at the pitch break, 24.5 ft at the ridge, all above the floor. The floor is 1′-1″ above the pad, so the eave is 15′-1″ above the pad, the ridge 25′-7″, and the mid-point of the gable 20′-4″. The gambrel slopes are 49° below the break and 16° above it. The county's pole barn guide, which does not bind an engineered building, puts its usual limits at a 14 ft side wall and a 27 ft peak, and the building stays inside both, measured from the floor (County rules asks where the county takes grade)
Orientation	Long side east–west; the main door is on the **east** gable, facing the paths and the RV pad, so a prevailing west wind meets the closed back wall. The baselines do not settle which way the winter wind blows
Design loads	120 psf ground snow (the county's zone for Prairie, and the load it required of a neighbor's building) · 115 mph wind, Exposure C · 40+ psf loft floor · seismic category set by the engineer
Main door	14′ × 12′ insulated, wind-rated sectional overhead door with low-headroom track, centered on the east gable. At 12 ft it clears the camper's roof by 1′-1″; at 14 ft wide it leaves 3′-0″ each side of the camper for backing a trailer in (a 12 ft door would leave 2 ft). The loft floor leaves 14″ above the door for the track
Loft doors	5′ × 6′ insulated double doors on the east gable, above the main door at loft-floor level, opening onto the deck
Hoist beam	W6 steel I-beam reaching 12′-6″ out through the east gable, 2 ft past the deck's outer edge, so a load lifts from the ground clear of the deck and comes in over the rail. Two raking braces rise from the deck's outer corners to the beam's tip. A 1-ton chain fall

	with a 25 ft lift, for loads of 500–1,000 lb
Landing deck	14′ × 10′, free-standing on 6×6 posts to grade, 42″ rail; reached from the loft only (no outside stairs)
Walk door	3′ × 7′ insulated steel, south wall near the east end
Windows	3 in the north wall and 3 in the south wall, 4′ × 3′ insulated sliders with the sill 5′ above the floor, centered in their bays and clear of the stair and the walk door; one 3′ × 4′ single-hung in the west gable at the loft. About $1,900 to $3,500 in materials with their flashing and trim
Loft access	L-shaped stairs of 24 risers of 7 in: the first flight (11 risers) along the south wall to a landing at the southwest corner 6′-5″ up, then the second flight (13 risers) up the west wall through a 4′ × 11′ opening in the loft floor
Heat and power	Wood stove and a mini-split heat pump (cooling too) · 200 A sub-panel; where it is fed from is still open
Colors	beige roof and trim · barn-red walls
Build approach	Built by the owner with family, with pre-fab stamped trusses, in phases if needed. The goal is weather-tight before snow flies
Budget	Materials about $79,300 to $122,800; about $103,600 to $176,700 all in, with tax, permits, engineering, utilities, rentals, and a 15% contingency (materials and cost)

Where It Sits

The workshop goes on the graded pad north of the pump house and the containers, with its east door lined up with the road-mix path and the RV path. Its footprint is in `data/site.json` at 32 × 48 ft, centered on the pad.

WHERE THE WORKSHOP GOES — THE PAD AND ITS SURROUNDINGS



The pad, the planned building, and what is around it, drawn from `data/site.json` on the 50 ft master grid. Distances are measured on the grid. Compare the homestead plot map.

Planned footprint	48 × 32 ft in `data/site.json` (feature `workshop`), long side east–west, the same 48 × 32 ft this package is drawn at
Center	**B18-E11-S13** (cell B18), x 61, y 863 · 43.473085, -115.594602
Corners	northwest A17-E37-S47 · northeast B17-E35-S47 · southwest A18-E37-S29 · southeast B18-E35-S29

Grid cells it touches	A17, B17, A18, B18
Pad	56 × 42 ft (`workshop_pad`, edges approximate), level. The building is centered on it, leaving 4 ft of pad on the west and east, and 5 ft on the north and south.
Fill on the pad	Imported fill, from the owner (October 2026): **almost 2 ft over the original ground along the north edge, 6 in to 1 ft along the south edge.**
Original ground under the footprint	The lidar, taken before the pad was filled, puts it between 4,757.6 and 4,759.1 ft: northwest 4,757.6, northeast 4,758.3, southwest 4,758.1, southeast 4,759.1. It rises toward the south and east, which fits the thicker fill on the north.
Pad surface (estimate)	About 4,759.7 ft: the original ground at the center (4,758.4) plus about 1.3 ft of fill. Shoot it with a level before anything is poured.
Neighbors	Pump house 28.5 ft south of the south wall · solar container 33.0 ft south · buried 1.5 in water line 11.0 ft south · water line along the west fence 16.4 ft west · RV pad 9.5 ft to the southeast · irrigation tank 38.0 ft south · planned house 53.5 ft to the southeast

Notes on the pad and the foundation

- **Fill and compaction.** Ask the engineer whether the fill was placed in compacted lifts. If that is not documented, plan to compact and test it, or to deepen the gravel base, before the slab goes on it.
- **The pier holes go through the fill.** The 60 in is measured from the pad surface, so the north row's holes pass through about 2 ft of fill and end about 3 ft into native soil, and the south row's end about 4 to 4½ ft into it. Each pier pad should bear on undisturbed native soil, or on fill the engineer accepts. Detail 3 and Section A draw this.
- **The north edge.** The fill is thickest along the north edge, where the pad reaches only 5 ft beyond the wall (4 ft past each end). Check that the edge is firm and not slumping, and keep the north wall's piers and the slab edge well inside it.
- **The pad surface.** The lidar predates the pad, so shoot the finished surface with a level and record it here. The slab, foam, and gravel stack up about 13 in above it, so the floor sits about 1′-1″ above the pad.

County Rules and What They Mean Here

From Elmore County's published Zoning and Development Ordinance (Title 7, chapters 1 and 2, adopted May 2018, amended 2023 and 2024), the building regulations (Title 9), the county's Notice of Proposed New Agricultural Building, and its Pole Barn/Post and Beam Standards (approved July 19, 2021), read in October 2026 at elmorecounty.org. The ordinance is being amended (case ZOA-2026-01, to match the updated Comprehensive Plan; the staff report for the October 22, 2026 hearing recommends continuing the review to November 18, 2026). The draft keeps the height table and the accessory- and agricultural-structure standards cited below, but it reorganizes chapter 2, so section numbers may change. This is a planning summary, not legal advice and not the county's answer: read the current text and get each answer in writing.

Rule	What the ordinance says	This building	Where it stands
Height limit of the zone Table 7-2-27 (A)	35 ft in the Ag, Rec, and RR districts. An agricultural structure in the Ag district may go to 80 ft.	Ridge 25′-7″ above the pad; mid-gable 20′-4″.	Within the limit. Confirm which district the parcel is in.
Height of an accessory building 7-2-42 D, and "building height" in Title 7 chapter 1	No taller than the house or 24 ft, whichever is greater. In a residential district, a building in the front yard may not be taller than the house at all. Height runs from the average ground at the front wall to the average height (mid-point) of the highest gable.	Mid-gable 20′-4″ above the pad surface, or 21′-8″ above the original ground (the fill averages 1.3 ft under the east wall).	Within the limit: under by 44″ measured from the pad, under by 28″ from the original ground, if the house is no taller than 24 ft. Averaging the gambrel's whole gable, not just halving eave and ridge, gives about 22.0 ft from the pad. Ask how the county reads a gambrel and where it takes grade.
Size of an accessory building 7-2-42 C	In the front yard, no more than 1,500 sq ft or half the house's footprint, whichever is less.	1,536 sq ft. The planned 40 × 66 ft house is 2,640 sq ft, so half is 1,320 sq ft.	Over the cap, if the building counts as in the front yard (between the road and the front of the house). Ask which road is the front. At 32 ft wide, a length of 46 ft (1,472 sq ft) would clear the 1,500 sq ft figure, and 41 ft (1,312 sq ft) would clear half the house too.
A house comes first 7-2-42 A	An accessory building needs a house on the same parcel with a certificate of occupancy or a valid building permit. It may not be a dwelling or be used for business.	The house is still planned.	Permit the house first, or be ready to show it under a valid permit.
Matching the house 7-2-42 E	Roofing and finish similar in color to the house, and a roof pitch similar to the house's. In a residential district, a front-yard building of 1,500 sq ft or more must also portray the house's architectural character.	Barn-red walls, a beige roof, and a gambrel (49° and 16°). At 1,536 sq ft it is over the size where the character rule starts.	Likely to be questioned. Ask how strictly the county applies it, or choose colors and a pitch that suit the house.
Agricultural building Title 9, 9-1-7; 7-2-46; the definitions in Title 7 chapter 1; the county's exemption form	Exempt from the building code and its fees, but not from zoning: it needs zoning approval first and the county's agricultural exemption, and the district's setbacks and height still apply. It must stand on a farm (5 acres or more in agricultural use) and be used **solely** to store farm implements, hay, grain, poultry, livestock, or horticultural products; the definition excludes personal items and a shop not related to agriculture. The owner signs a notarized statement, recorded with the county, and the county may inspect within 12 months to confirm the use is still agricultural.	Drawn as an RV garage, a workshop, and a storage loft.	Does not fit as drawn: an RV and a general workshop are personal use. A barn for orchard equipment, a tractor, bins, and harvest storage would fit if the parcel counts as a farm, and then the plans and the signed statement must describe that use. A later change of use may be a violation.

Rule	What the ordinance says	This building	Where it stands
Snow load and the engineer Pole Barn/Post and Beam Standards, pages 1 and 3; the county's building permit page	The county's guide, which lets a simple pole barn skip an engineer, is for the 30 psf snow zone and 115 mph wind only, and a 120 psf area always needs engineering. The permit page says the 120 psf zone must be designed by an Idaho engineer. The guide also allows no habitation, wants a loft kept open (no rooms), and sends any deviation from its standard method to a registered design professional.	120 psf ground snow: the county's map puts Prairie in that zone, and the county required a neighbor's building to be designed for it (owner, October 2026). 115 mph wind. A 32 × 48 ft gambrel (1.50 to 1, where the guide asks for at least 5 to 3) with an open storage loft, a hoist, and a free-standing deck.	The guide does not apply: at 120 psf this is an engineered building. An Idaho engineer stamps the plans and the trusses, which are specified and priced for 120 psf. The next row gives the guide's usual wall and peak limits, for reference.
Wall and peak in the pole barn guide Pole Barn/Post and Beam Standards (July 19, 2021), page 1 and sheet 1	A side wall over 14 ft needs engineering, and the peak may not exceed 27 ft in total height. An alternate design stamped by an Idaho design professional may be considered by the Building Official.	Side wall (the eave): 14 ft above the floor, 15′-1″ above the pad surface. Roof peak: 26′-4″ above the pad surface, which is the top of the trusses (24.5 ft above the floor) plus about 9 in of purlins, deck, steel, and ridge cap.	The wall limit only decides whether a barn needs an engineer, and this one has one. The peak limit is a stated maximum that a stamped design may pass only if the Building Official accepts it. The standards do not say where either is measured from. From the floor the wall is 14′-0″, and from the pad surface 15′-1″. The roof peak is 8 in under 27 ft from the pad surface, so extra fill or slab thickness would use that margin up, and 8 in over 27 ft from the original ground, which is 1.3 ft lower under the east wall. Ask the county where it takes grade. The 14 ft wall is what the 12 ft camper door and its track need under the loft floor.

Which permit path is this building on?

The county sorts a building by what it is used for, not by its name.

- **Accessory building: the design as drawn.** An RV garage, a workshop, and a storage loft. The house comes first, and the height, size, and matching rules above apply. It needs a building permit, plan review, and inspections. At 120 psf the county's pole barn guide, which lets a simple barn skip the engineer, does not apply (see the table), so the plans and the trusses need an Idaho engineer's stamp.
- **Agricultural building.** A barn for orchard equipment, a tractor, bins, and harvest storage. It is exempt from the building code, but not from zoning: it needs zoning approval before construction and the county's agricultural exemption in place of a building permit, and the district's setbacks and height still apply. It must stand on a farm (5 acres or more in agricultural use, tied to the assessor's tax categories), so ask whether the parcel qualifies. The owner signs a notarized statement, recorded with the county, that the building is used solely for agriculture.

The two cannot be mixed: the definition says *solely*. With the RV inside, the building is on the first path, and a building later put to another use can be required to obtain every permit and prove it meets the code. If the plan changes to a farm barn, the package should be redrawn for that use (a different name, no camper clearances), and the application should describe what the building will hold.

Before Any Site Work

1. **Call the Elmore County Land Use and Building Department** (208-587-2130, extension 6500): confirm the wind speed, frost depth, and exposure for this parcel, and which rules apply at about 4,750 ft. The ground snow load is settled at 120 psf: the county required a neighbor's building to be designed for it.
2. **Settle the permit path with the county, in writing** (see County rules): the parcel's zoning district (some of the rules apply only in residential districts); whether this is an accessory building or an agricultural building, which follows from what it will really hold; how the height is measured, from what grade (for the pole barn guide's 14 ft side wall and 27 ft peak as well as the 24 ft accessory limit), and how a gambrel is averaged; the front-yard size cap; and whether the house must be permitted first.
3. **Get stamped truss drawings** for 120 psf ground snow from a pre-fab truss company: send it the truss package, which has the collar tie, the loft stair opening, the loft doors, and the hoist-beam loads on it.
4. **Get stamped foundation and structural plans** from an Idaho-licensed engineer for 120 psf ground snow and 115 mph wind, including what to do about the pad's fill.
5. **Pull the building permit.**
6. **Call 811** before any digging. The buried 1.5 in water line is about 11 ft south of the south wall and the west-fence line about 16 ft west.
7. **Settle the power source** and run the sub-feed conduit before backfill.
8. **Shoot the pad's elevation** with a level and get the fill checked before the footings.

Open Questions

1. **What will the building really hold?** That decides the permit path (see County rules). An RV garage and a workshop make it an accessory building, which is what these drawings show. A barn for orchard equipment and harvest could be an agricultural building. It cannot be both.
2. **Where does the power come from?** The pump house with its solar inverters and batteries, or the house? A 200 A panel is a panel rating, not a supply: check that the source can run the welder (50 A), the compressor (30 A), and the mini-split together.
3. **How heavy a load will the hoist lift?** The beam is sized for 1,000 lb at its tip, with the two braces carrying that load down into the deck. A heavier load, such as an engine or a full pallet, changes the beam, the braces, and the deck's piers, so decide what it will lift before the engineer sizes them.

4. **How far does the slab slope?** The slab slopes toward the east door. At ¼ in per ft over 48 ft that is 12 in, which is a lot. Slope only the 12 to 16 ft nearest the door, or use ⅛ in per ft; decide before the forms are set.
5. **Stove, condenser, and panel.** The wood stove's spot is not chosen. The mini-split condenser goes on the west or the south wall.
6. **How will the inside walls be insulated and finished?** The girts are on the outside of the posts, so between the posts there is nothing to fasten insulation or an interior finish to. A conditioned shop needs horizontal strapping across the posts (2×4 laid flat at 24 in OC, about 1,120 ft in all) or a framed inner wall, with batts in the 7¼ in bays. The materials estimate prices the batts and the finish sheets but not the strapping, so settle this before the second phase is priced.
7. **The winter wind.** The door is on the east end on the idea that a prevailing west wind hits the closed back wall. If the winter wind comes from the east instead, the door and the deck take it and the drifts pile up there. The weather station's first year should settle it.
8. **Camper height.** The drawings use the manufacturer's listed height, 10′-11″. That leaves 1′-1″ under the 12 ft door head and 2′-3″ under the loft floor. Measure it as it will be parked, hitched, and check that the figure takes in the roof air conditioner and anything else up there. A taller door needs a taller wall and longer posts, and a wall over the pole barn guide's usual 14 ft is the engineer's call, since this building is engineered anyway.
9. **How does the county apply the 14 ft wall and the 27 ft peak to an engineered building?** The pole barn guide states them for a barn without an engineer, and lets the Building Official consider a stamped alternate design. It does not say where it measures them from. From the floor the wall is 14′-0″; from the pad surface it is 15′-1″. The roof peak, with about 9 in of purlins, deck, steel, and ridge cap on the trusses, is about 26′-4″ above the pad, 8 in under 27 ft. Measured from the original ground, which is about 1.3 ft lower under the east wall, it would be about 8 in over.
10. **Is the upper roof slope steep enough?** It is 3.3 in 12 (16°), kept low so the roof peak stays under the county's usual 27 ft. Ask the steel supplier for the minimum slope of its exposed-fastener panel (many ask for at least 3 in 12, or sealant at the laps below that), and ask the engineer how much snow drifts at the pitch break at 120 psf on a slope this flat, and whether a steeper slope is worth a taller peak.
11. **How will the roof be insulated?** The truss package keeps the loft's room open between the pitch-break lines, crossing it only with the collar ties (8′-0″ up, on T1 to T20). A ceiling on those ties would be flat across only the middle 13′-6″ of the loft, so blown-in insulation on a flat ceiling cannot cover the roof. The insulation has to follow the roof instead: spray foam against the deck, or batts between the top chords with a vent channel above them, with the 12 in heel giving it depth at the eaves. Decide the method with the engineer and the truss plant, and price it: the materials estimate carries blown-in cellulose on a flat ceiling as a placeholder.

How to Use This Package

1. Verify the design loads and the county rules (see County rules) with Elmore County before any orders go out.
2. Use the specs when calling the steel vendor or the concrete supplier; read straight off the page. Send the truss package (or its PDF) to the truss plant: it holds the shape, the open room and its collar tie, the loads, and the special trusses.
3. Use the plans and elevations for layout on site and with the permit; the framing in perspective shows how the posts, girts, trusses, and purlins fit together, and the wall framing sheets show every girt board.
4. Use the sections and details when framing, sheathing, and siding; the numbered and lettered markers match the legends.
5. Use the materials list as the shopping and check-off list. Firm prices replace the estimates as quotes come in.
6. Build the posts from the post schedule: three full-length 2×8s, 30d nails, two rows on the 7¼ in face. It is a fastening schedule, not the engineer's stamp.
7. Dig the piers from the footing page: mini excavator, 60 in below the pad, a solid 30 in pad in a ring form, and an open 18 in tube set on that pad, poured together, backfilled the next day.

Sources

- Site records: site master grid and homestead plot map (`data/site.json`: the planned workshop, its pad, the paths, the water lines, and the neighbors), and the USGS lidar terrain.
- County building values: site baselines (Elmore County Land Use and Building Department: snow zone, frost depth, wind, and seismic zone).
- The design snow load: 120 psf, the county's snow zone for Prairie, and the load the county required a neighbor's building to be designed for (reported by the owner, October 23, 2026).
- County zoning and building rules: Elmore County Zoning and Development Ordinance, Title 7 chapters 1 and 2 (definitions; table 7-2-27; sections 7-2-42 and 7-2-46), Title 9 section 9-1-7, the Notice of Proposed New Agricultural Building with its exemption statement (elmorecounty.org, October 2026), and the staff report and draft text for ZOA-2026-01 (October 22, 2026 hearing).
- County pole barn guide: Elmore County Pole Barn/Post and Beam Standards, approved July 19, 2021 (for the 30 psf snow zone only, which is why it does not apply here: its 14 ft side wall, its 27 ft peak, and the engineering it requires at 120 psf).
- The camper: the manufacturer's specifications for the 2024 Forest River Rockwood Mini Lite 2517S (length, width, height, and weights), from the owner, October 23, 2026.
- The pad's fill, level, and thickness: the owner, October 23, 2026.

Mountain Prairie Acres · Workshop pole barn overview · updated October 23, 2026 · every figure is drawn from `docs/workshop_model.js`, and `tools/test_workshop_model.js` checks it against `data/site.json` and the county values in `data/site_baselines.json`.